

Lab 6: System Identification

Section 1

Summary

Give a summary in your own words of what you did in this lab. You can talk about why it is important? What did you learn? What were the challenges? How did this contribute to the overall S1XT33N project?

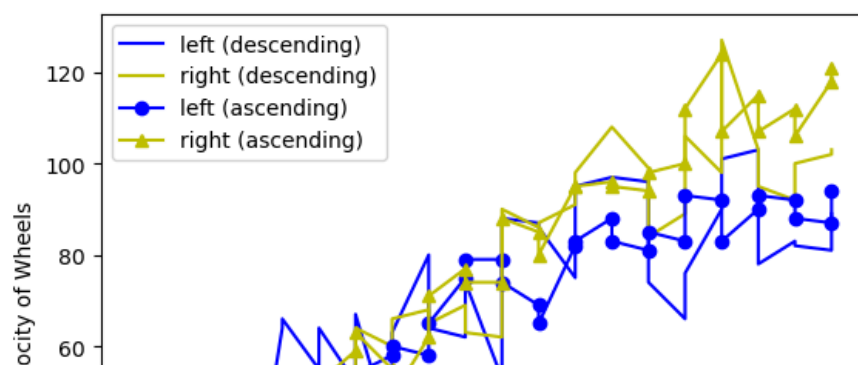
- We selected a PWM input range for fine data collection based on where the left and right wheels overlapped the most during coarse data collection, optimizing our data collection process. Additionally, we discussed the potential benefits and drawbacks of implementing a higher order model compared to a linear model, considering the trade-offs between accuracy and complexity. We also explored how decreasing battery levels affect the car's performance, predicting a linear decrease in velocity due to lower voltage output, contributing to our understanding of the car's behavior in real-world scenarios within the S1XT33N project.

1. What do Θ and $\square$ represent physically, not mathematically? What are your values of Θ and $\square$, and do they reflect the car's performance while collecting data? Why/why not?

- Theta (Θ) represents the change in input PWM to change in velocity while beta ($\square$) represents the velocity offset that encompasses real world imperfections like static friction. Our values of Θ_{left} is 0.3811 and Θ_{right} is 0.4371 while $\square_{\text{left}}$ is -16.8 and $\square_{\text{right}}$ is -9.645, they reflected the car performance as the car kept moving the right. One wheel turned faster than the other.

2. How did you choose the PWM input range for fine data collection? If there were other data ranges that could have also conceivably been chosen, why did you choose this range over other ranges? Please include a graph of your car's coarse data to support your answers.

We chose the PWM input range based on where the left and right wheels overlapped the most during coarse data collection. There were other values we could've chosen with some overlap, but we picked a range of around 50 PWM values where the left and right wheels were closest together. We picked the range of 100 - 150 PWM.



3. To implement a higher order polynomial model, what would you need to change in the current lab flow to calculate the coefficients for this new model? Evaluate the benefits and drawbacks between using a higher order model vs. a linear model.

To implement a higher order polynomial model, we would have to use a non-linear regression model. The benefit of that would be that we may be able to get a better fit line to our data with a lower error value. However, this comes with the drawback of higher complexity and more time spent learning how to do a new regression model.

4. As the batteries run low, assume that the corresponding velocities for each PWM input go down linearly. In other words, the velocity vs. PWM input curves for both wheels shift down by some value, where v_0 is non-negative and increases as the battery power gets lower. If we still aim for the same operating point velocity and use the same linear model, how will the car's performance change as the battery level decreases?

- As the battery level decreases, the relation between each PWM and velocity input curves will remain linear, however, the car's performance would change in speed since the voltage that is being sent out for each duty cycle is lower; it would run slower. We would then be using the same operating point system and adjusting/approximating for a linear system determined with a higher input voltage.

Lab 7: Controls

Section 2

Summary

Give a summary in your own words of what you did in this lab. You can talk about why it is important? What did you learn? What were the challenges? How did this contribute to the overall S1XT33N project?

- In this lab, we implemented open and closed control systems. We adjusted PWM inputs based on feedback to regulate wheel velocities, aiming for straight-line trajectories and controlled 90 degree turns. We also noted for steady-state error correction and jolt application to overcome static friction. To maintain stability while using positive f-values and adjusted for negative f-values to model corrections for trajectory in the opposite direction.

Questions

1. What are the open loop equations of S1XT33N's control scheme for our PWM input, $u[i]$? What are some advantages and disadvantages of open loop control?

- The open loop equations for $u[i]$ are

- $u_L[i] = 1/\theta_L(v^* + \beta_L)$
- $u_R[i] = 1/\theta_R(v^* + \beta_R)$
- An advantage of these equations is that since this is in an ideal scenario, you wouldn't need to periodically adjust for $u_L[i]$ and $u_R[i]$, whereas there are predetermined inputs for the system model. It also doesn't require feedback measurements. You would also have both V_L and V_R both as the target velocities = V^*
- A disadvantage of these equations is that open loop doesn't consider any type of feedback which includes disturbances such as noise, friction, or any real world scenarios that would cause interference. Ultimately preventing the car from driving in a straight trajectory or adapting to external factors.

2. What are our equations for closed loop control? Why might we want to incorporate closed loop control, rather than open loop, in SIXT33N?

- The closed loop equations are where $\delta[i] = d_L[i] - d_R[i]$:
- $u_L[i] = 1/\theta_L(u_L^{OL} - (f_L/\theta_L)\delta[i])$, where $u_L^{OL} = 1/\theta_L(v^* + \beta_L)$
- $u_R[i] = 1/\theta_R(u_R^{OL} + (f_R/\theta_R)\delta[i])$, where $u_R^{OL} = 1/\theta_R(v^* + \beta_R)$
- We would want to implement closed loop control rather than open loop control because we would need to accurately adjust for wheel velocities based on the differences in distance traveled between left and right wheels $\delta[i] = d_L[i] - d_R[i]$. Additionally, we would adjust for real world disturbances such as any noise to maintain a straight trajectory.

3. What is our system eigenvalue? What are the conditions on f_L and f_R such that our system eigenvalue is internally stable?

- Our system's eigenvalue is $(1 - f_L - f_R)$. The conditions on f_L and f_R such that our system's eigenvalue is internally stable is that the magnitude of the eigenvalue must be less than 1, this is to make sure our discrete time system bounds are "bounded" and do not diverge over time.

4. What is the effect of setting both $f_L = 0$ and $f_R = 0$?

- Setting both f_L and f_R to 0 would result in the system operating in open-loop control for no feedback which would lead to predetermined inputs. This would ultimately be faulty since we are not accounting for external factors which would lead to inaccuracies in motion control for a straight car trajectory.

5. What is the purpose of the jolts? Why might we have different jolts for left and right wheels?

- The purpose of the jolts are to overcome static friction in order to get the car to move. Different jolts for the left and right motor parameters account for armature resistances, along with difference in motor efficiencies of a high to low ratio of friction and lastly accounts for any mass imbalances in the wheels of the car.

6. Why can't we use negative f-values for both wheels? If we wanted to use negative f-values for both wheels, how should we change our closed-loop model equations such that our car goes straight and corrects any errors in its trajectory?

- We can't use negative f-values for both wheels because it would result in an unstable closed-loop system, leading to unbounded behavior. If we wanted to adjust for this and use negative f-values, we would need to change our close loop model equations by adjusting the sign of the feedback term (δ) for correcting trajectory in the opposite direction.

7. What does a zero delta ss value tell you about your car's trajectory? What about a non-zero delta ss value? What kind of error is it supposed to correct when we add it to our control scheme? (Hint: Think about the difference between the trajectories for a zero versus a non-zero delta ss value.)

- A zero δ_{ss} value indicates a steady straight trajectory for our car which would drive straight. A non-zero δ_{ss} value indicates that there needs to be adjustments to wheel distances to correct for maintaining a straight-line motion.

8. How did you change the closed-loop model equations to allow the car to turn? Write the equations below and explain how they change for turning left, turning right, and going straight.

- For turning left, the δ_{ref} term would be added to simulate a left turn, while for turning right, a negative δ_{ref} term would be added to simulate a right turn. The equation we derived for δ_{ref} in terms of r, v* for going straight is $\delta_{ref}[i] = l(v^*)(i)/r$

9. Describe how the trajectory of the car would look if we had a constant δ_{ref} , rather than our δ_{ref} , which changes as a function of the timestep.

- A constant δ_{ref} compared to a $\delta_{ref}[i]$ would result in a consistent turning behavior throughout the trajectory, leading to a circular curve for the motion of the car. However, for a function $\delta_{ref}[i]$ of the timestep would allow for control over turning radius and velocity parameters to prompt controlled 90 degrees left and right turns.

Lab 8: Controls

Section 3

Summary

Give a summary in your own words of what you did in this lab. You can talk about why it is important? What did you learn? What were the challenges? How did this contribute to the overall S1XT33N project?

- In this lab, voice recordings were collected and tested to identify the most distinguishable data for four specific commands: left, right, long straight, and short straight. Through

testing and elimination of words prone to being mistaken for noise, a final selection of four words was made, optimizing the classifier's accuracy. Technical methods like SVD/PCA were used to compress data and enable classification on the memory constrained Arduino.

Questions

1. What 4 words did you choose for classification? What characteristics of this set make your words good for classification? Provide at least two features. Compared to other sets of words with similar ideally good characteristics, why is this set preferable to others?

- The four words chosen for classification are "Latte," "Smoothie," "Pancake," and "Bread." The characteristics that made these words good for classification were the distinct number of syllables needed to reduce misclassification and difference in pronunciation stress patterns. Some had a stretch in pronunciation such as pancake and smoothie and the other two words had different ending syllables.

2. Why is taking the envelope of your voice signals a good choice for classification, especially considering that this classifier is implemented on an Arduino?

- It significantly reduces the noises and complexity that comes with taking samples of each word, we took about 40 samples and then averaged them for PCA construction to get out top 3 principal components. This process is extremely important for the Arduino because it has low memory/storage and overloading the Arduino with many sample vectors of data compared to loading it with our PCA vectors of sampled data would be more efficient, essential for classification.

3. Why do we need to use SVD/PCA to represent our data set?

- We need SVD/PCA to be used to represent the dataset because it compresses the data to a manageable size, important for the memory constraints of the Arduino. PCA also identifies the principal components capturing the most significant variations while reducing disturbances such as noise in the dataset which is crucial for classification and to develop our data clusters.

4. If we were to use the transpose of our data matrix for SVD, which rows/columns of which matrix correspond to the principal component vectors representing the recorded words? Why?

- If we were to use the transpose of the data matrix for SVD (V^T), the principal component vectors representing the recorded words would correspond to the rows of matrix U. Since matrix U contains the column vectors representing the principal components when the original data matrix is used.

5. How many basis vectors are you using, and how did you choose this number? What are the benefits vs. tradeoffs of increasing or decreasing the number of basis vectors by a small amount? What about increasing the number of basis vectors by a large amount?

- We used 3 basis vectors because it was advised by lab staff to use about 3 for the memory constraints in our Arduino. The benefits vs. tradeoffs were increasing the

number up to 3 would significantly improve accuracy of our sample data but any other increase aside from 3 would overload the Arduino vs decreasing the number of basis vectors would capture much data.

6. What are length, prelength, and threshold for our data pre-processing? How does changing them affect your alignment? Include both the definitions and the values you chose. What kinds of words are better suited for our pre-processing method with live classification?

- To begin with, we interpreted the threshold as the maximum amplitude in the recording that defines when a word starts and from our clusters, the data closest to our radius around the centroids that capture most of the data. The length is the total number of samples that our word occupies in each recording. The prelength is the number of samples to include before the start of the word as determined by the threshold. Changing them would affect our alignment by needing to correct for the segment of the word in term of classification accuracy. Our Euclidean threshold was 0.04 and our loudness threshold was 140.

7. Why can we simply take the dot product when projecting our recorded data vector onto the principal component vectors?

- The dot product simplifies projection because it measures the similarity between vectors, aligning with the goal of identifying patterns in the data. It reduces computational complexity compared to other projection methods, making it suitable for implementation on resource-constrained devices like the Arduino.

8. What is EUCLIDEAN THRESHOLD? What is LOUDNESS THRESHOLD? Include both the definitions and the values you chose. During live classification, which threshold was more difficult to satisfy, and how do you know?

- Our Euclidean Threshold: 0.04 the maximum distance to centroid for valid word classification. Our Loudness Threshold: 140, the minimum sample value for considering it as a word. During live classification adjusting for these thresholds was challenging due to variations in spoken words compared to recorded data which was really observed through misclassifications and retesting.

9. How different was live classification in practice from what you found in the SVD/PCA lab? Why do you think that is?

- The live classification was different from SVD/PCA lab due to real-world factors like background noise such as other groups testing their words at same time as ours and variations in spoken words. These factors significantly increased the complexity of classification, leading to discrepancies between expected and observed results. We adapted by adjusting thresholds and loudness thresholds.

Feedback

Section 4

1. Extra Credit: To receive extra credit, please provide 1-2 sentences for each of the following parts:

- a) How well do you feel that this assignment evaluated your understanding of labs this semester? I felt as though going back and answering lab specific questions post lab was always extremely helpful to solidify my understanding of what I just did in the lab. Reading the pre-labs would often feel hazy to me as though sometimes I would have to do the following task where I had to envision components and scenarios in the car that were unfamiliar to me. During the lab section I would complete tasks but sometimes the bigger picture wouldn't come full circle until I answered post lab questions.
- b) How much time do you think this assignment took you to complete? (Just a number is fine for this part.) 5 hours
- c) What changes to lab reports would improve your experience? The changes I would make to the lab report to improve my experience fewer questions that are motivated by driving the bigger picture.

2. Additionally, please feel free to provide any feedback you have about 16B lab or anything we can do to better support you.

- I think dividing labs up into 2 weeks is the best way to fully understand lab concepts and to ensure proper functionality of the car. This would have taken care of groups who may have experienced mishaps such as burnt out batteries, dysfunctional components, etc... and given them the proper time to have every aspect of the car good to go. This would also let other groups advance but keep a steady state in terms of completion.

Collaborators/ resources

- Lab staff